HWA CHONG INSTITUTION
JC2 Preliminary Examination
Higher 2

CANDIDATE NAME		**CT GROUP**	15S ___
CENTRE NUMBER		**INDEX NUMBER**	

BIOLOGY **9648 / 02**

Paper 2 Core Paper **13 September 2016**

Additional Materials: Writing Paper **2 hours**

INSTRUCTIONS TO CANDIDATES

Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page.

SECTION A

This section contains **eight** questions. Answer **all** questions.
Write your answers on the lines / in the spaces provided.

SECTION B

This section contains **two** questions. Answer any **one** question.
Your answers must be in continuous prose, where appropriate.
Write your answers on the writing paper provided.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

A **NIL RETURN** is required for parts not answered.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
Question	**Marks**
1	**/ 8**
2	**/ 8**
3	**/ 11**
4	**/ 8**
5	**/ 10**
6	**/ 9**
7	**/ 12**
8	**/ 14**
9 / 10	**/ 20**
Total	**/ 100**

BOOKLET 1

SECTION A: STRUCTURED QUESTIONS

QUESTION 1

Proteins play an important role in many biological processes.

(a) **(i)** State **two** secondary structures commonly found in proteins.

.... [1]

(ii) Compare the secondary structures stated in **(a)(i)**.

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.... [2]

Fig. 1.1 shows a mammalian DNA polymerase interacting with a DNA molecule. The catalytic and binding amino acid residues of the DNA polymerase are located at different positions on a single polypeptide chain. These residues are brought close together in the active site.

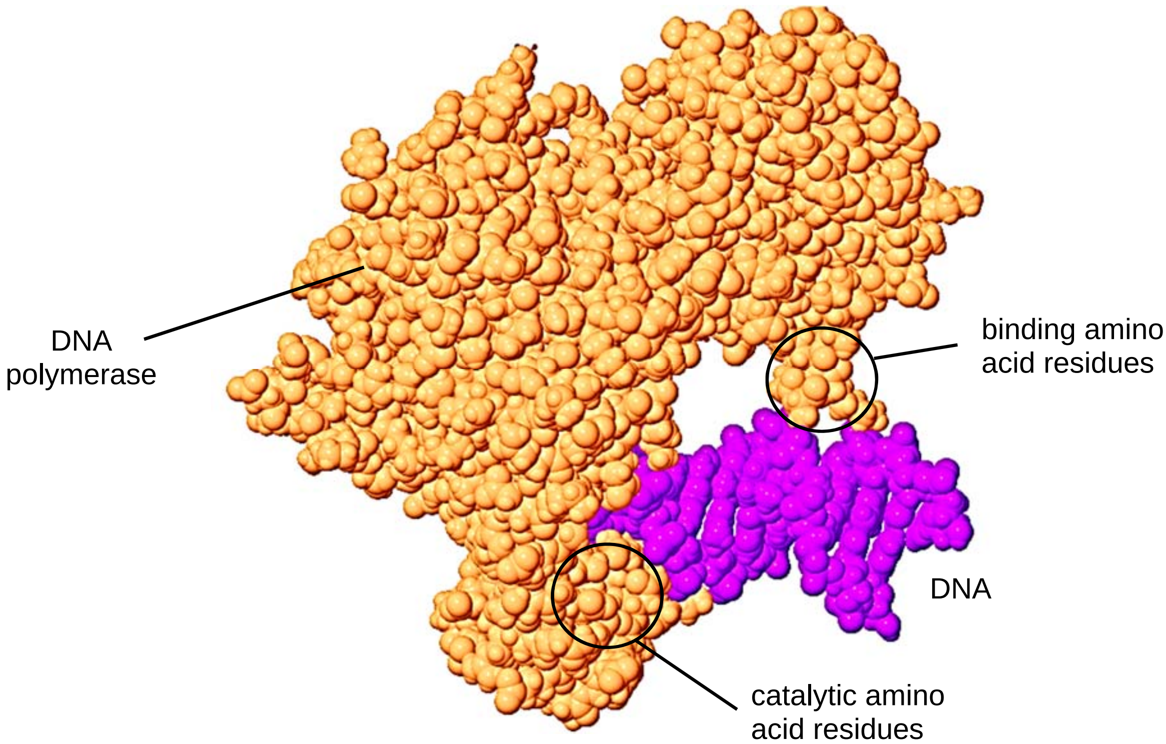

Fig. 1.1

(b) Describe how the catalytic and binding amino acid residues of DNA polymerase located at different positions are brought close together in the active site.

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.... [3]

Taq DNA polymerase has a similar function to the mammalian DNA polymerase. It is involved in polymerase chain reaction (PCR) in the presence of a pH 8.4 buffer.

(c) Explain the significance of the pH 8.4 buffer in PCR.

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.... [2]

[Total: 8]

QUESTION 2

Fig. 2.1 shows information about the movement of chromatids in a cell that has just started metaphase of mitosis.

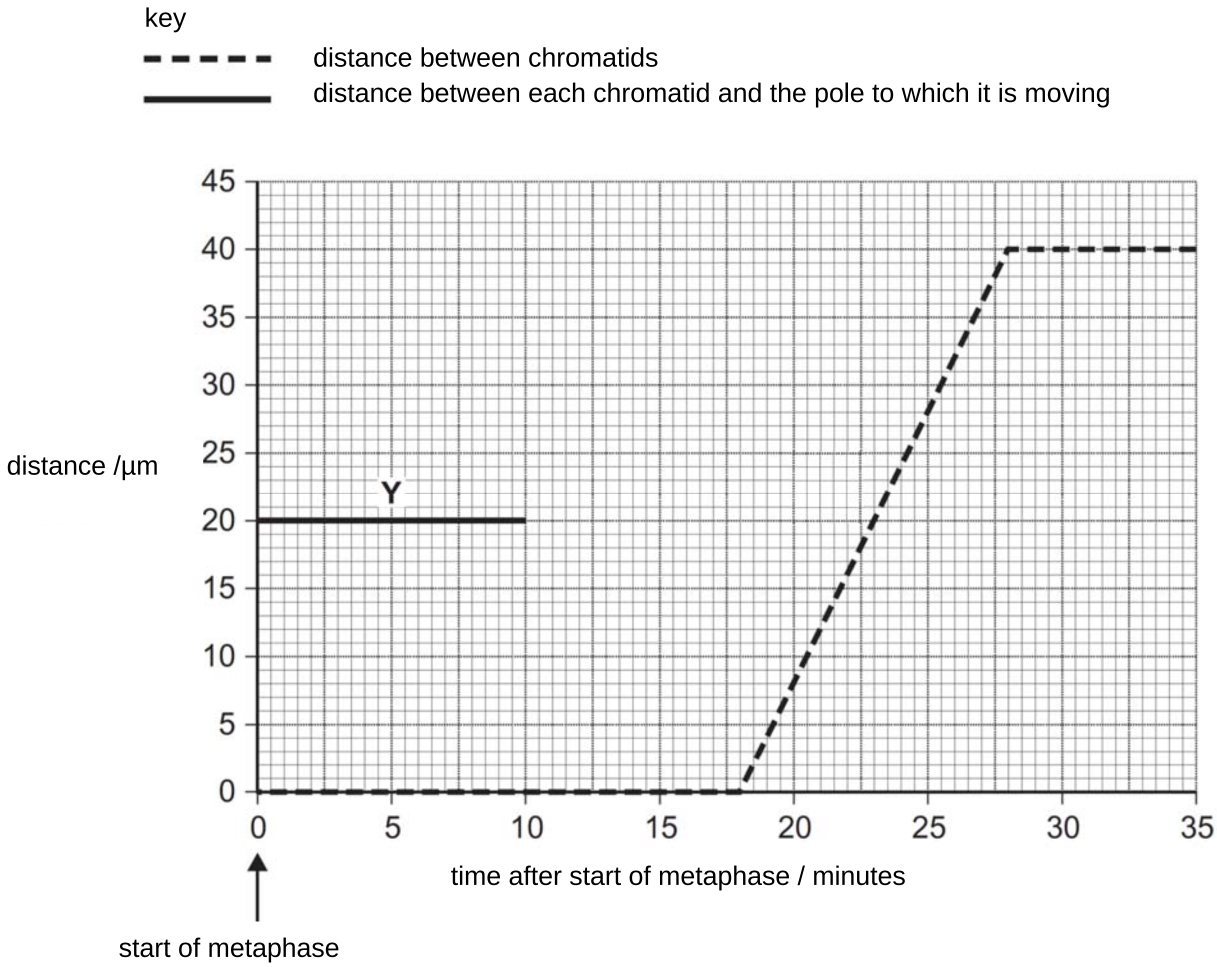

Fig. 2.1

With reference to Fig. 2.1,

(a) **(i)** state the duration of metaphase in the cell.

.... [1]

(ii) complete line **Y** on the graph. [1]

(iii) account for your answer in **(a)(ii)**.

[3]

The movement of chromatids is dependent on spindle fibres, which are made up of many tubulin subunits. Spindle fibres are lengthened at one end during mitosis by the polymerisation of tubulin subunits through GTP hydrolysis.

A drug, eribulin, is known to prevent the polymerisation of the tubulin subunits.

(b) Suggest and explain the effect of eribulin on the behaviour of chromosomes in mitosis.

[3]

[Total: 8]

QUESTION 3

p53 protein is a transcriptional activator that regulates the expression of *Mdm2* gene. Mdm2 protein in turn regulates the activity of p53 protein on genes involved in growth arrest, DNA repair and apoptosis.

Fig. 3.1 shows how p53 protein activates *Mdm2* gene expression.

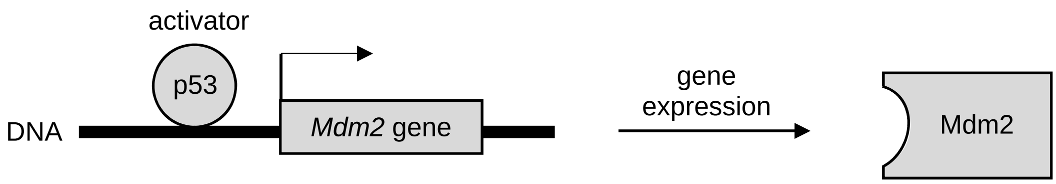

Fig. 3.1

(a) Explain the significance of p53 protein in the regulation of *Mdm2* gene expression.

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.... [3]

The cellular level of p53 protein is tightly regulated by Mdm2 protein.

When no DNA damage is detected in a cell, unphosphorylated p53 protein can bind to Mdm2 protein to trigger degradation of p53 protein via the ubiquitin system as shown in Fig. 3.2a.

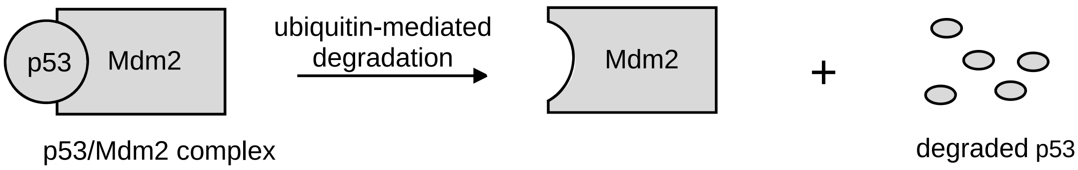

Fig. 3.2a

When DNA damage is detected in a cell, p53 protein will be phosphorylated at Serine15, Threonine18 or Serine20 amino acid residues as shown in Fig. 3.2b. In normal cells, these three amino acid residues are not phosphorylated, and p53 protein is maintained at low levels by Mdm2 protein.

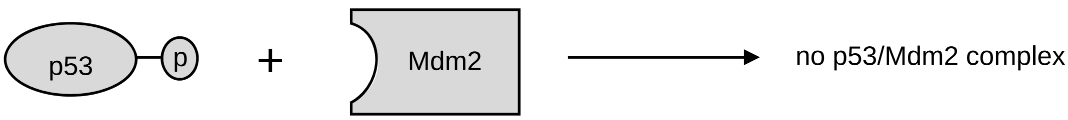

Fig. 3.2b

(b) Account for the high levels of p53 protein when DNA damage is detected in the cell.

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.... [4]

(c) Suggest the need for ubiquitin-mediated degradation of p53 protein to occur.

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.... [1]

(d) Homozygous deletion of *Mdm2* gene in mouse germline cells results in lethality at the blastocyst stage, due to inappropriate apoptosis.

With reference to Fig. 3.2a and Fig. 3.2b, suggest how inappropriate apoptosis occurs.

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.... [3]

[11 marks]

QUESTION 4

The retrovirus, human immunodeficiency virus (HIV), and the influenza virus are two types of enveloped viruses. Both enter the human host cells by adsorption and penetration. Fig. 4.1 shows the entry process of a HIV into a macrophage, which is a type of white blood cell.

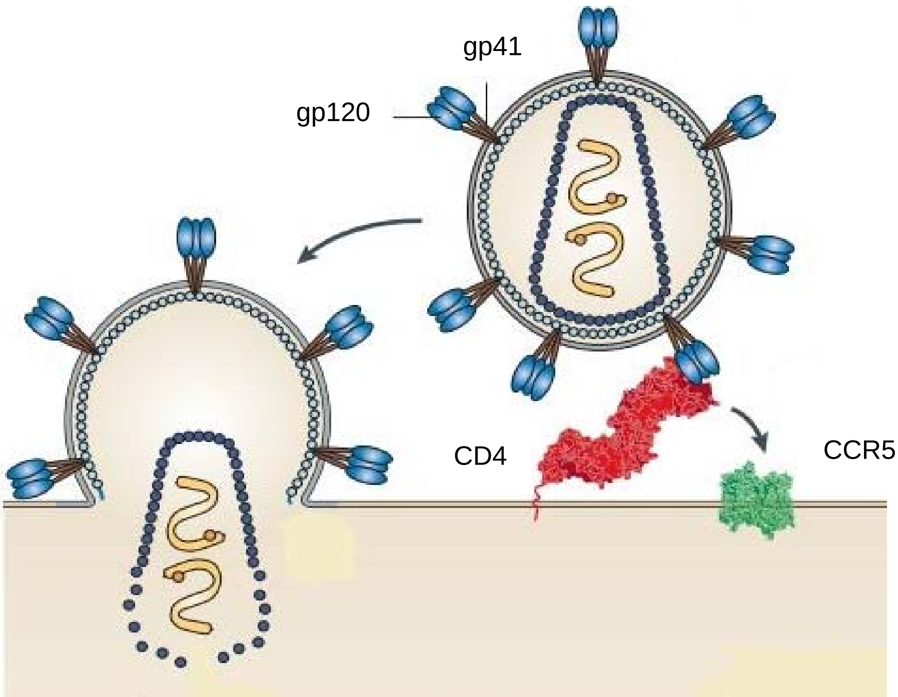

Fig. 4.1

(a) **(i)** State what is meant by *retrovirus*.

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.... [2]

(ii) Compare the entry processes of the HIV and influenza virus into human host cells.

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.... [3]

(b) Upon completion of the entry process, describe how the genome of HIV is inherited.

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.... [3]

[Total: 8]

QUESTION 5

George Schull, a botanist at Princeton University, conducted a genetic study of a common weed known as shepherd's purse, *Capsella bursa-pastoris*. He studied the shape of its fruit, which could be heart-shaped or narrow respectively, as shown in Fig. 5.1.

Fig. 5.1

When he crossed a pure-breeding plant with heart-shaped fruit to a pure-breeding plant with narrow fruit, the F1 generation all had heart-shaped fruit. When the F1 generation was self-fertilised, the numbers of F2 generation with the respective shaped fruit were recorded as follows:

heart-shaped fruit	2251
narrow fruit	150

Fig. 5.2 shows the possible biochemical pathway that results in the observation made in the F2 generation.

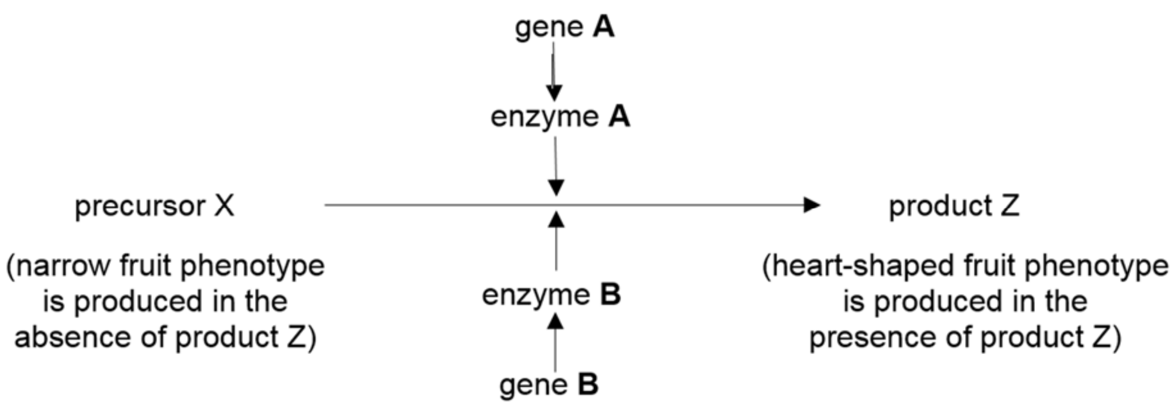

Fig. 5.2

(a) Explain the type of gene interaction observed in this context.

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.... [3]

(b) Using the symbols **A**, **a**, **B** and **b**, draw a genetic diagram to explain the results of the F2 generation in the space provided. [5]

(c) Describe how you could identify *Capsella* sp. plants with heart-shaped fruits, which are homozygous dominant in at least one gene locus.

....

....

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.... [2]

[Total: 10]

QUESTION 6

The unicellular green alga, *Chlorella*, a photosynthetic eukaryote is mass produced and harvested by commercial suppliers for use as a health food supplement.

Fig. 6.1 shows the effect of carbon dioxide concentration on the light-independent stage of photosynthesis in *Chlorella*. The following steps were carried out in a study:

- a cell suspension of *Chlorella* was illuminated using a bench lamp.
- the suspension was supplied with carbon dioxide at a concentration of 1% for 200 seconds.
- the concentration of carbon dioxide was then reduced to 0.03% for a further 200 seconds.
- the concentration of RuBP and glycerate-3-phosphate (GP) were measured at regular intervals.
- the temperature of the suspension was maintained at 25 °C throughout the investigation.

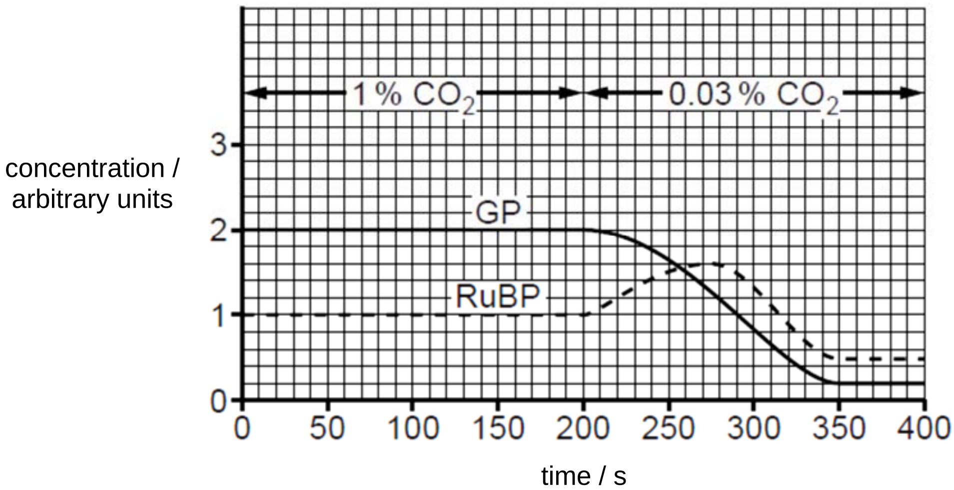

Fig. 6.1

(a) **(i)** State precisely where RuBP and GP are produced in the chloroplast.

.... [1]

(ii) Explain why the concentration of RuBP changed between 200 and 275 seconds.

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.... [2]

(b) Suggest how the decrease in the concentration of GP leads to a decrease in harvest for commercial suppliers of *Chlorella*.

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.... [2]

(c) In the light dependent stage, illumination of chloroplasts is important for maintaining the high pH in the stroma.

Explain how the illumination of chloroplasts maintains the high pH in the stroma.

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.... [3]

(d) The endosymbiotic theory postulates that the chloroplasts of *Chlorella* evolved from bacteria living within an eukaryotic host cell.

Suggest **one** structural similarity between the chloroplasts and the bacteria that supports this theory.

....

.... [1]

[Total: 9]

QUESTION 7

Fig. 7.1 shows part of a myelinated motor neurone. The numbers show the membrane potential, in millivolts (mV), at various points along the axon of the myelinated neurone.

(a)(i)

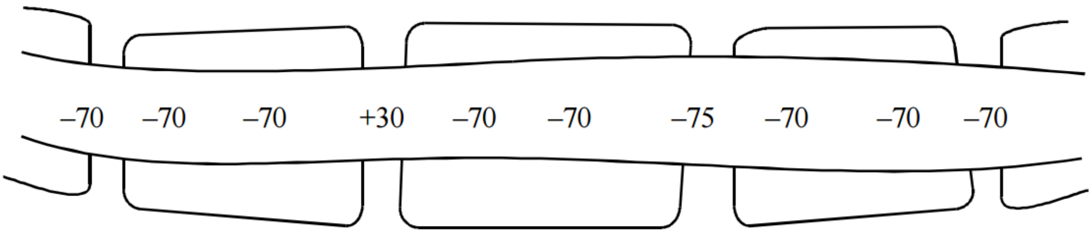

Fig. 7.1

(a) **(i)** Draw an arrow in the box provided on Fig. 7.1 to indicate the direction in which one nerve impulse is being conducted. [1]

(ii) Explain your answer in **(a)(i)**.

....

....

....

.... [2]

(b) Suggest why transmission of nerve impulses along a myelinated neurone uses less energy in the form of ATP than transmission along an unmyelinated neurone.

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.... [2]

Fig. 7.2 shows the changes in the concentration of sodium ions inside a segment of an axon. The neurone was stimulated at the points indicated on the graph. The neurone was treated with dinitrophenol (DNP) in region **Y**.

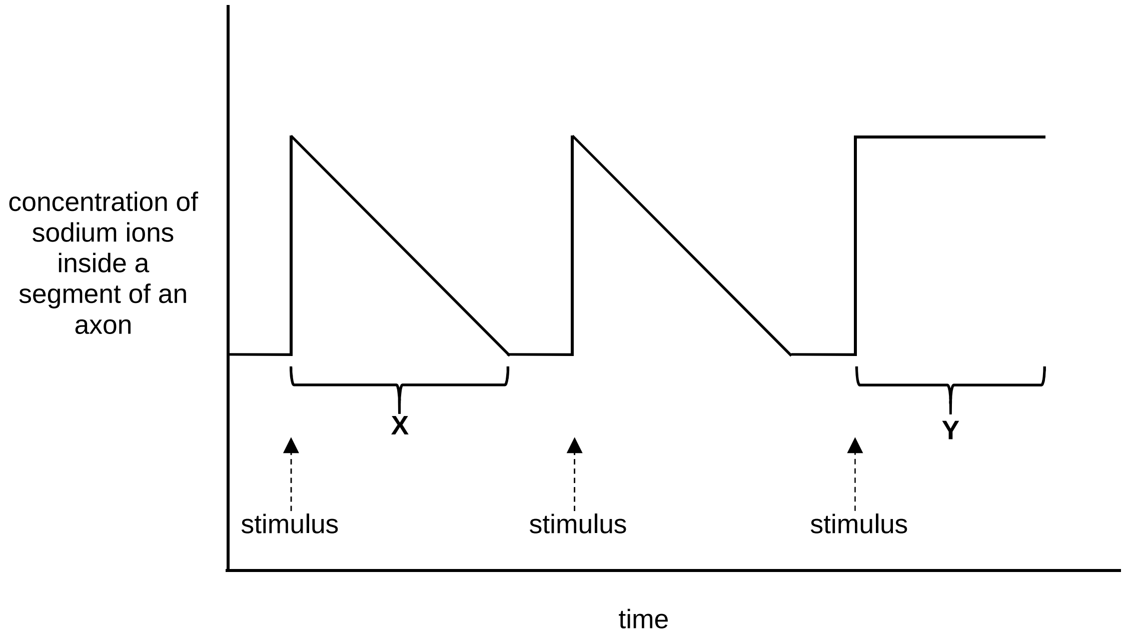

Fig. 7.2

(c) Account for the changes in the concentration of sodium ions at region **X** upon the stimulation of the neurone.

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.... [4]

DNP, a metabolic poison, makes the inner mitochondrial membrane leaky to protons during aerobic respiration.

(d) Suggest and explain why the concentration of sodium ions remain constant at region **Y**.

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.... [3]

[Total: 12]

QUESTION 8

Whales are marine mammals that belong to the order Cetartiodactyla. For a long time, scientists believed that whales are related to pigs and hippopotamuses. To reconstruct the evolutionary relationships between whales, pigs and hippopotamuses, scientists studied their limb and inner-ear bones as shown in Table 8.1.

Table 8.1

type of animal	limb bones	inner ear bones
whales	thick	thick
pigs	thin	thin
hippopotamuses	thick	thin

(a) Explain how the anatomical similarities among the whales, pigs and hippopotamuses support Darwin's theory of natural selection.

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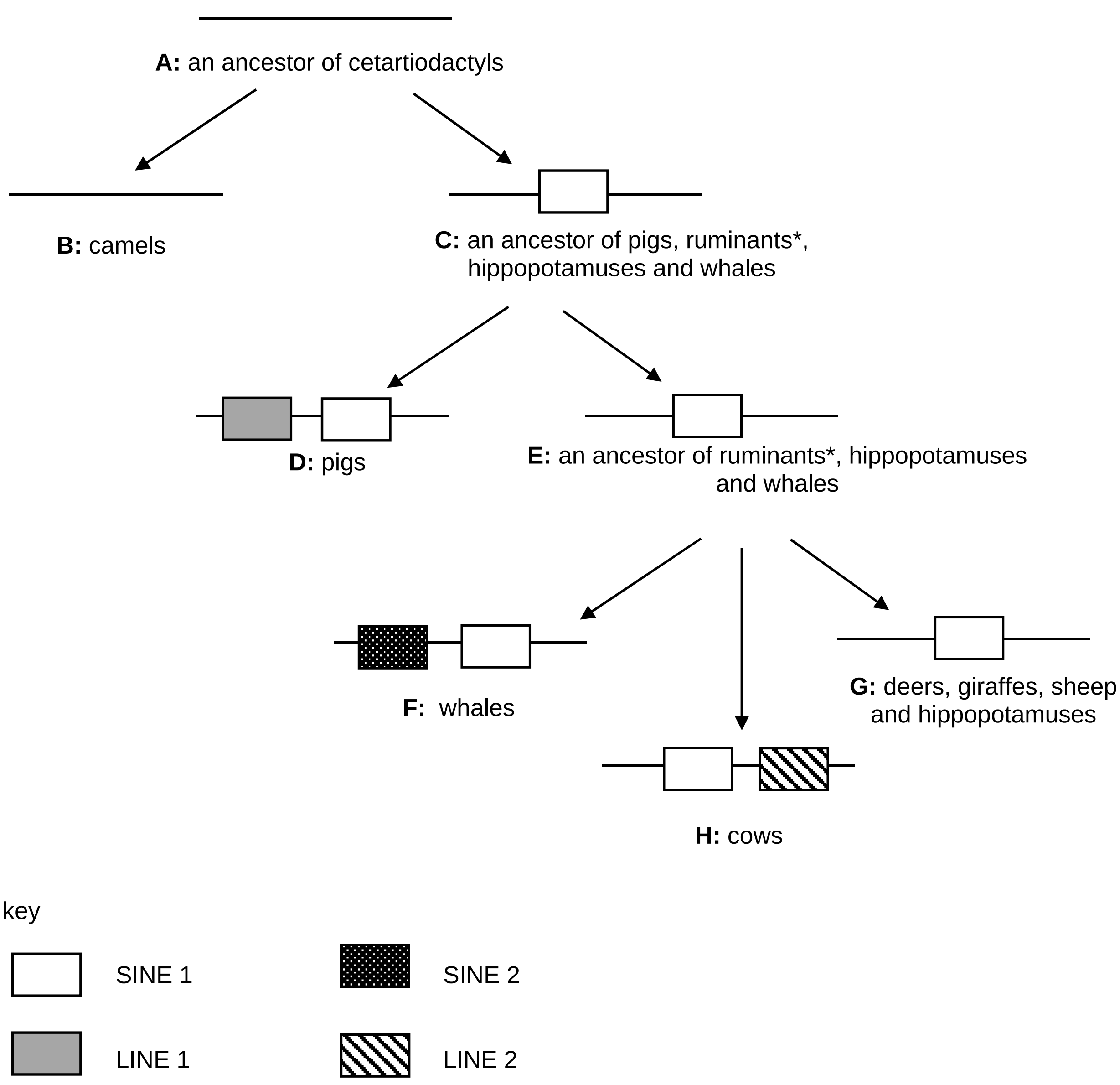
.... [3]

Phylogenetic relationships can be elucidated using molecular data. An important method of analysis involves the study of both Short Interspersed Elements (SINEs) and Long Interspersed Elements (LINEs). SINEs and LINEs are non-coding DNA sequences that have been amplified and inserted into different genomic regions.

Fig 8.1 is a schematic representation of SINEs and LINEs insertions at homologous genomic regions among the subgroups of cetartiodactyls.



*ruminants include deers, giraffes, sheep and cows

Fig. 8.1

(b) **(i)** Explain why camels and whales are of different species according to the phylogenetic species concept.

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.... [3]

(ii) A phylogenetic tree is constructed based on the results of SINEs and LINEs analysis in Fig 8.1. Fill in the blanks at the nodes and end points of the phylogenetic tree with the corresponding letters, **A** to **H** as represented in Fig. 8.1. [2]

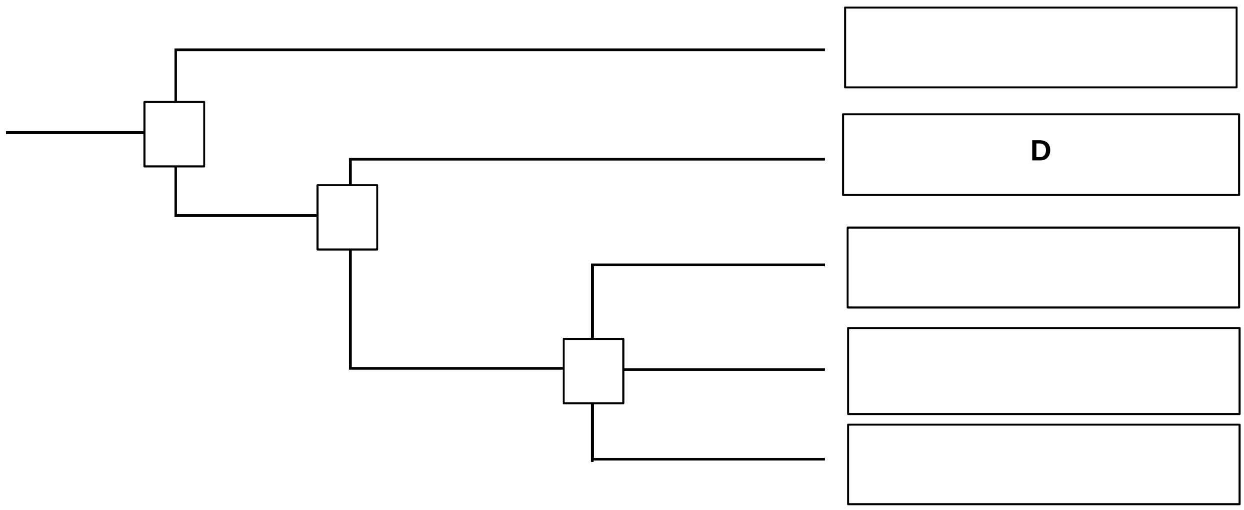

Fig. 8.2

(c) Molecular changes in SINEs and LINEs occur over time due to mutations and these are hypothesised to follow the neutral theory of evolution.

Briefly describe the neutral theory of molecular evolution and suggest how it may be used to study the subgroups of cetartiodactyls.

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.... [4]

(d) The presence of SINEs and LINEs increase the genome size over time. However, this does not contribute to a significant increase in nuclear size.

Outline how the organisation of the genome allows for DNA packaging in a non-dividing cell.

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.... [2]

[Total: 14]

--- End of Section A ---

SECTION B: FREE RESPONSE QUESTION

Answer **one** question.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
You answer must be set out in parts **(a)**, **(b)** etc., as indicated in the question
A **NIL RETURN** is required for any parts not answered.

QUESTION 9

(a) Describe how the molecular structure of cellulose is related to its function. [6]

(b) Outline the basis of the selective permeability of the cell membrane with reference to phospholipids, cholesterol and proteins. [8]

(c) Explain why animal cells mainly store lipids instead of carbohydrates. [6]

[Total: 20]

QUESTION 10

(a) Describe how the molecular structure of the G-protein coupled receptor is suited for its role in glucagon-mediated cell signalling. [8]

(b) Outline the concept of negative feedback in regulating glucagon levels in the body. [6]

(c) Explain how signal amplification is illustrated upon the binding of insulin to its receptor. [6]

[Total: 20]

--- End of Section B ---

--- End of Paper ---